

W2 – Ratios for Angles Greater than 90°

MCR3U

1) Sketch each angle in standard position and determine the related acute angle.

a) 135°

b) 210°

c) 315°

d) -30°

e) -225°

f) -330°

g) 150°

h) -120°

i) 105°

j) -163°

k) -141°

l) -280°

2) State the value of each ratio **exactly**.

a) $\sin 225^\circ$

c) $\sin 270^\circ$

b) $\cos 240^\circ$

d) $\tan 300^\circ$

e) $\cos 180^\circ$

3) Point P (-9, 4) is on the terminal arm of an angle in standard position.

a) Sketch the principal angle, θ .

b) What is the measure of β , the related acute angle to the nearest degree?

c) What is the measure of θ to the nearest degree?

4) Point P (7, -24) is on the terminal arm of an angle in standard position.

a) Sketch the principal angle, θ .

b) What is the measure of β , the related acute angle to the nearest degree?

c) What is the measure of θ to the nearest degree?

Answers

1) a) 45° b) 30° c) 45° d) 30° e) 45° f) 30° g) 30° h) 60° i) 75° j) 17° k) 39° l) 80°

2) a) $-\frac{1}{\sqrt{2}}$ b) $-\frac{1}{2}$ c) -1 d) $-\sqrt{3}$ e) -1

3) b) 24° c) 156°

4) b) 74° c) 286°